Edit

Guided

Explore Document Processing in Azure AI's Document Intelligence Studio

Challenge Overview

Understand the scenario

You are a Developer for Hexelo, an organization that needs to create a Custom extraction model by using Azure AI Document Intelligence Studio.

In this Challenge Lab, you will explore document processing in Azure AI's Document Intelligence Studio. First, you will set up and explore the Document Intelligence Studio. Next, you will extract and process different types of data, and then you will create a Custom extraction model. Finally, you will train a Custom extraction model.

Navigating the Challenge Lab

▶ Understanding a Cloud Slice

▶ Quick tips for navigating the Challenge Lab instructions.

New to Challenge Labs? Click here to learn more.

Set up and explore Document Intelligence Studio

Hints Enabled

No Yes

Note About Evolving Technology

Azure AI technologies are constantly evolving as Microsoft continues to update, refine, and improve its tools and interfaces. As a result, you may notice slight differences between the instructions in this lab and what you see on your screen. Rest assured, we are actively working to keep all tutorials up to date with the latest features and interfaces. We appreciate your patience and understanding, and we apologize for any inconvenience this may cause.

- Sign into the virtual machine as **Admin** using T Passw0rd! as the password.
- Open **Microsoft Edge**, go to T https://portal.azure.com, sign in to the Microsoft Azure portal as T {USERNAME} using T {PASSWORD} as the password, and then dismiss all prompts.

Expand this hint for guidance on signing into the Azure portal.

- On the Microsoft Windows taskbar, select the **Microsoft Edge** button.
- In the Sign in dialog, enter T {USERNAME}, and then select **Next**.
- In Enter password, enter T {PASSWORD}, and then select **Sign in**.
- Dismiss all prompts.

Want to learn more? Review the documentation on the Azure portal.

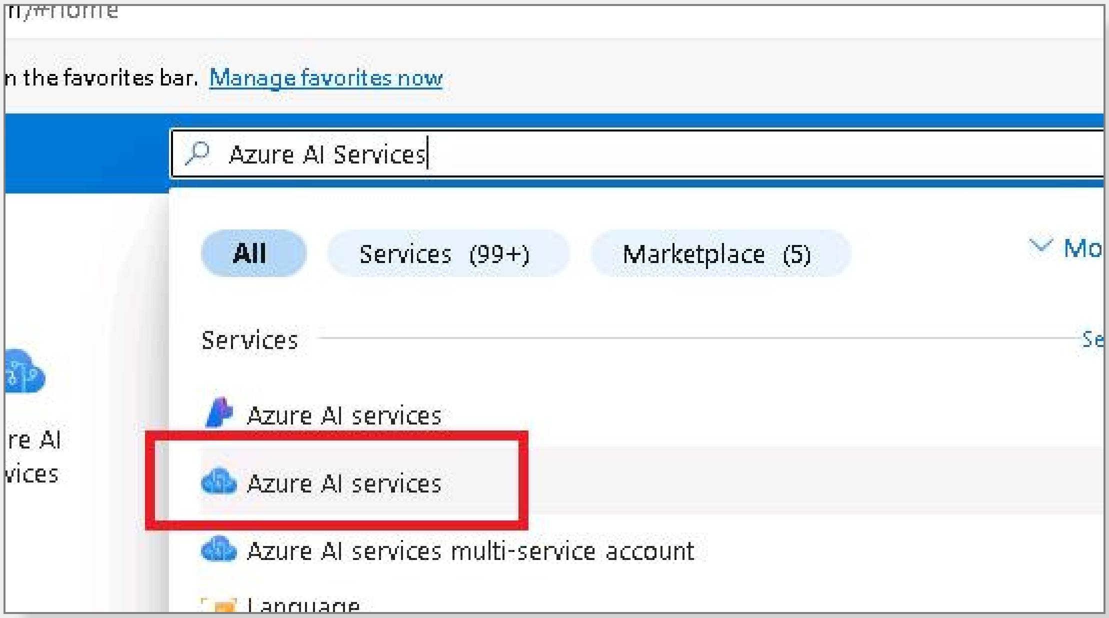
- Create a new T Azure AI Document intelligence service named T Hexelo-Doc-Intel-{LAB_INSTANCE_ID} by using the values in the following table. For any property that is not specified, use the default value.

Property	Value
Resource group	**{RESOURCE_GROUP_NAME}**
Region	**East US**

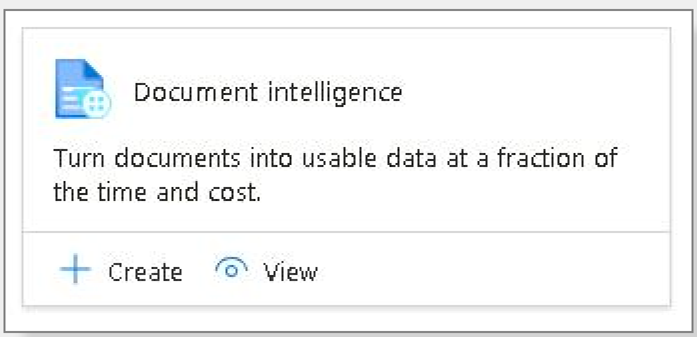
Property	Value
Name	T Hexelo-Doc-Intel-{LAB_INSTANCE_ID}
Pricing Tier	**Free F0**

Expand this hint for guidance on creating a new Azure AI document Intelligence service.

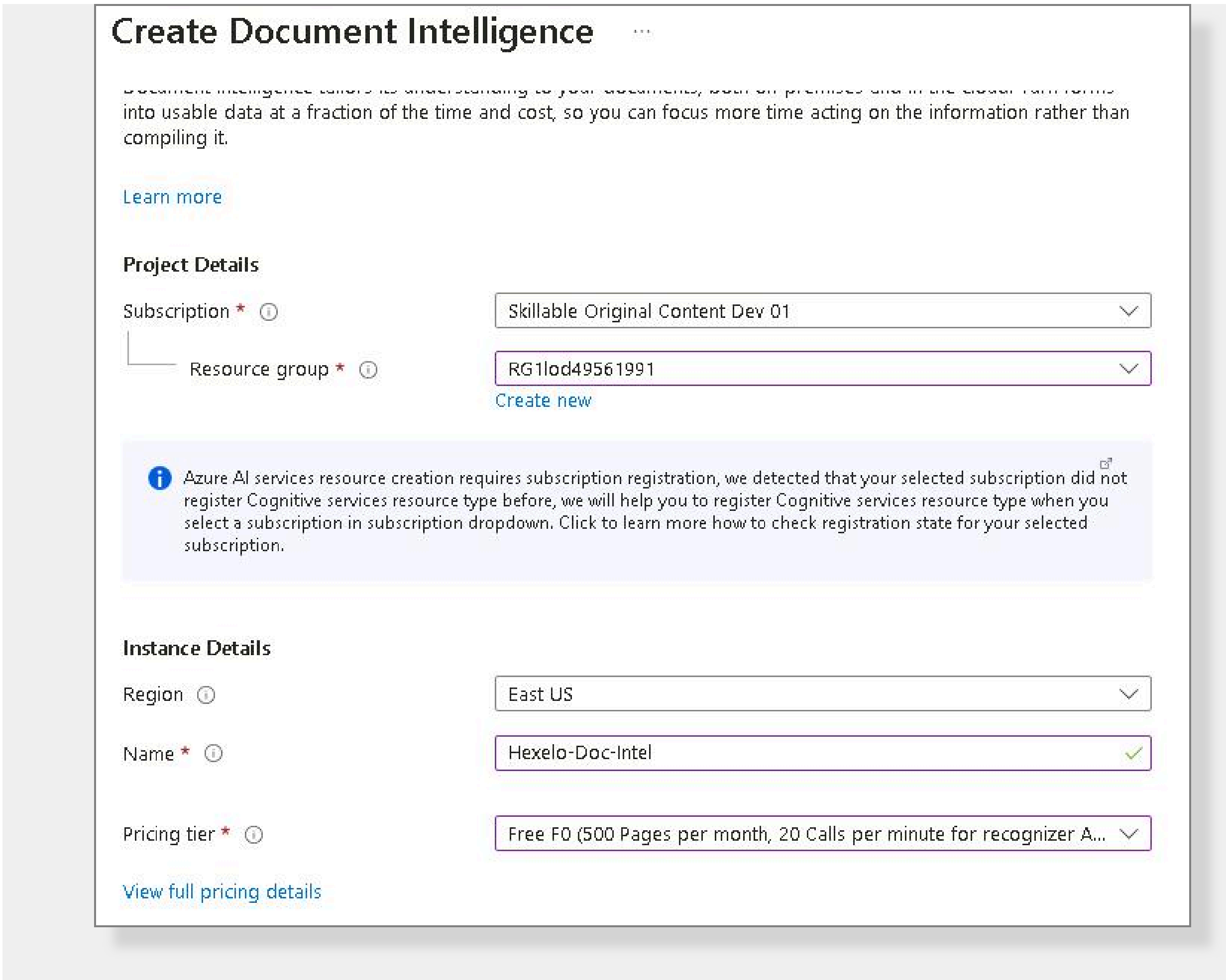
- On the Microsoft Azure global toolbar, search for and select T Azure AI Services.



- On the Azure AI services service menu, in Azure AI services, select **Document intelligence**.
- On the Document intelligence command bar, select **Create**.



- On the Create Document Intelligence blade, on the Basics page, in Project Details, in Resource group, select **RG1-{LAB_INSTANCE_ID}**.
- In Instance Details, in Name, enter T Hexelo-Doc-Intel-{LAB_INSTANCE_ID}.
- In Pricing tier, select **Free F0**.
- On the Create Document Intelligence blade, select **Review + create**, and then select **Create**.



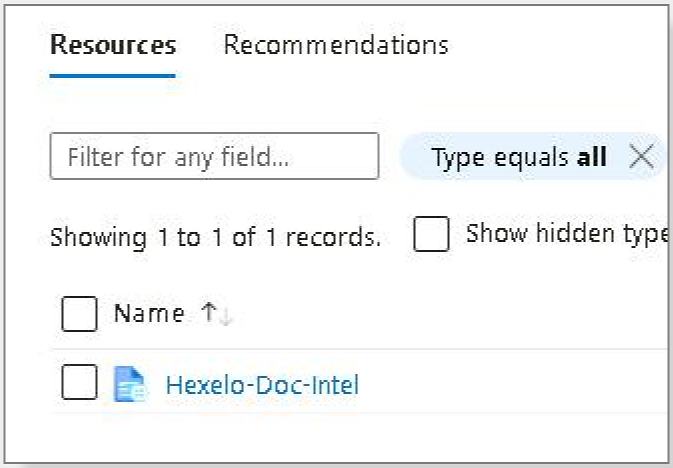
Want to learn more? Review the documentation on creating a new Azure AI document intelligence service.

- Record the **KEY 1** API Key, in the following **KEY 1** text box:

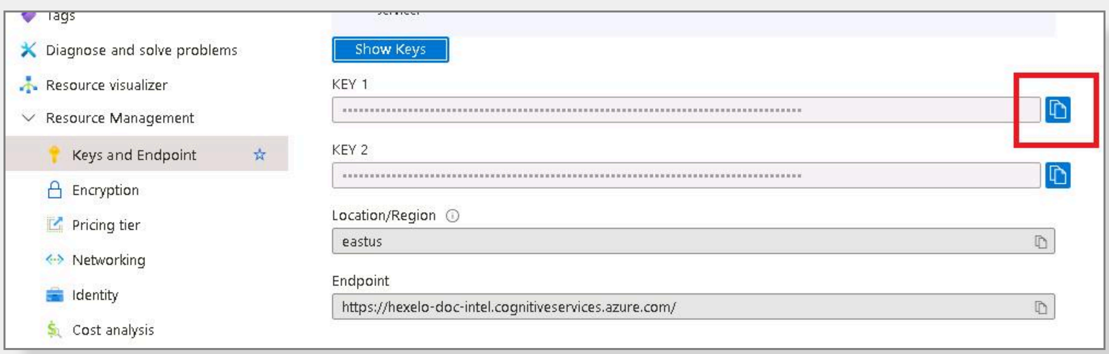
KEY 1

Expand this hint for guidance on locating the API Key.

- On the Deployment page, select **Go to resource**.
- On the RG1-{LAB_INSTANCE_ID} blade, in Resources, select **Hexelo-Doc-Intel-{LAB_INSTANCE_ID}**.



- On the Hexelo-Doc-Intel-{LAB_INSTANCE_ID} service menu, in Resource management, select **Keys and Endpoint**.
- On the Keys and Endpoint page, select the copy-to-clipboard icon to the right of **KEY 1**, and then paste the copied KEY into the **KEY 1** text box in the Challenge Lab instructions.

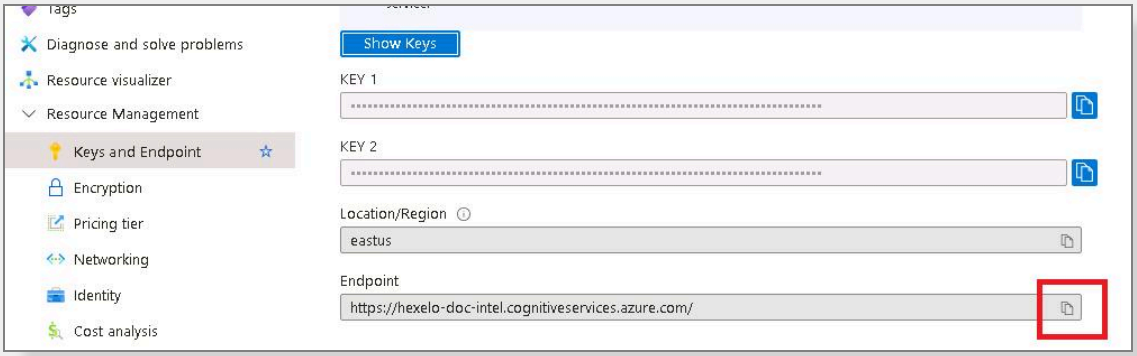


- Record the **Endpoint** URL in the following **Endpoint URL** text box:

Endpoint URL

Expand this hint for guidance on locating the Endpoint URL.

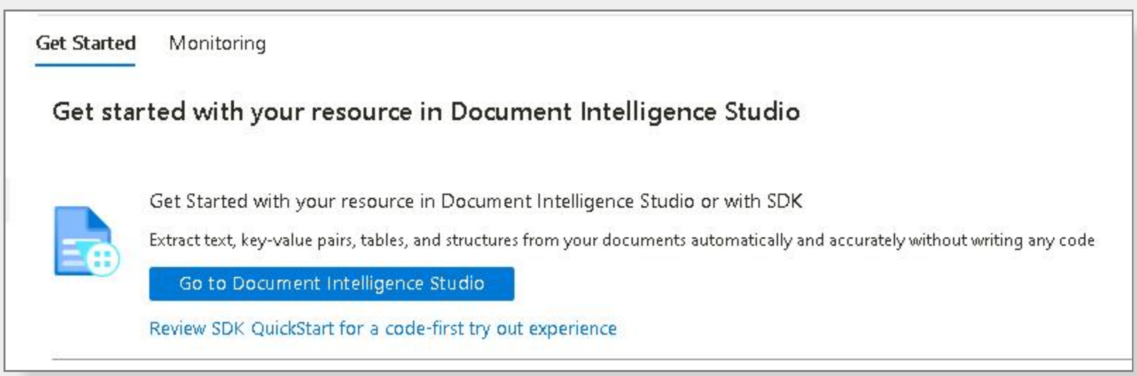
- On the Keys and Endpoint page, select the copy-to-clipboard button to the right of **Endpoint**, and then paste the copied Endpoint into the **Endpoint** text box in the Challenge Lab instructions.



- Access **Document Intelligence Studio**.

Expand this hint for guidance on accessing the Document Intelligence Studio.

- On the Hexelo-Doc-Intel-{LAB_INSTANCE_ID} service menu, select **Overview**.
- On the Overview page, in Getting started with your resource in Document Intelligence Studio, select **Go to Document Intelligence Studio**.



Want to learn more? Review the documentation on accessing the Document Intelligence Studio.

Check your work

Verify

Extract and process different types of data

Hints Enabled

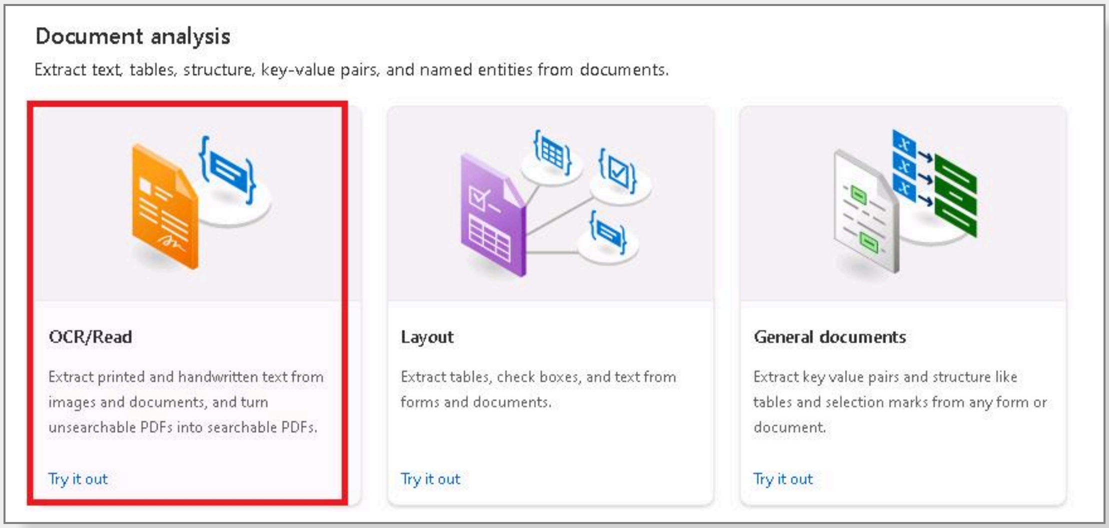
No Yes

- Configure a service resource for OCR/Read document analysis that uses the Default subscription-for example, Skillable Original Content Dev01-the **{RESOURCE_GROUP_NAME}** Resource group, and **Hexelo-Doc-Intel-{LAB_INSTANCE_ID}** as the *Document Intelligence or Cognitive Service Resource*.

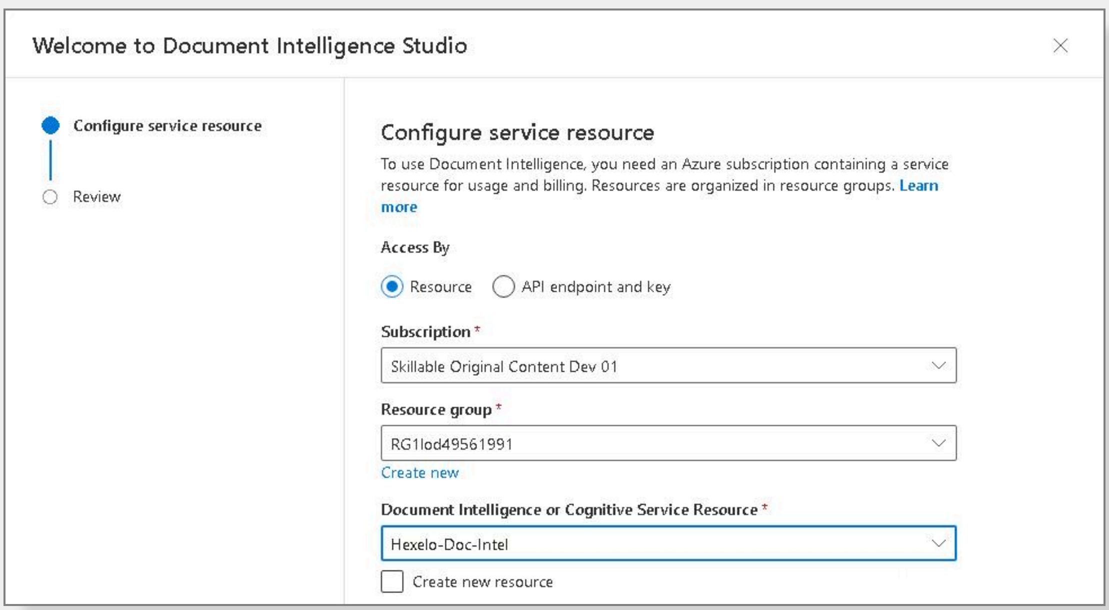
If prompted, sign in as T {USERNAME} using T {PASSWORD} as the password.

Expand this hint for guidance on configuring a service resource.

- On the Get Started with Document Intelligence Studio page, in Document analysis, in **OCR/Read**, select **Try it out**.



- If needed, sign in as T {USERNAME} using T {PASSWORD} as the password.
- In the Welcome to Document Intelligence Studio dialog, in Configure service resource, in Access by, ensure that **Resource** is selected.
- In Subscription, select the default subscription-for example, Skillable Original Content Dev01.
- In Resource group, select **{RESOURCE_GROUP_NAME}**.
- In Document Intelligence or Cognitive Service Resource, select **Hexelo-Doc-Intel-{LAB_INSTANCE_ID}**, and then select **Continue**.



- Review your selections, and then select **Finish**.

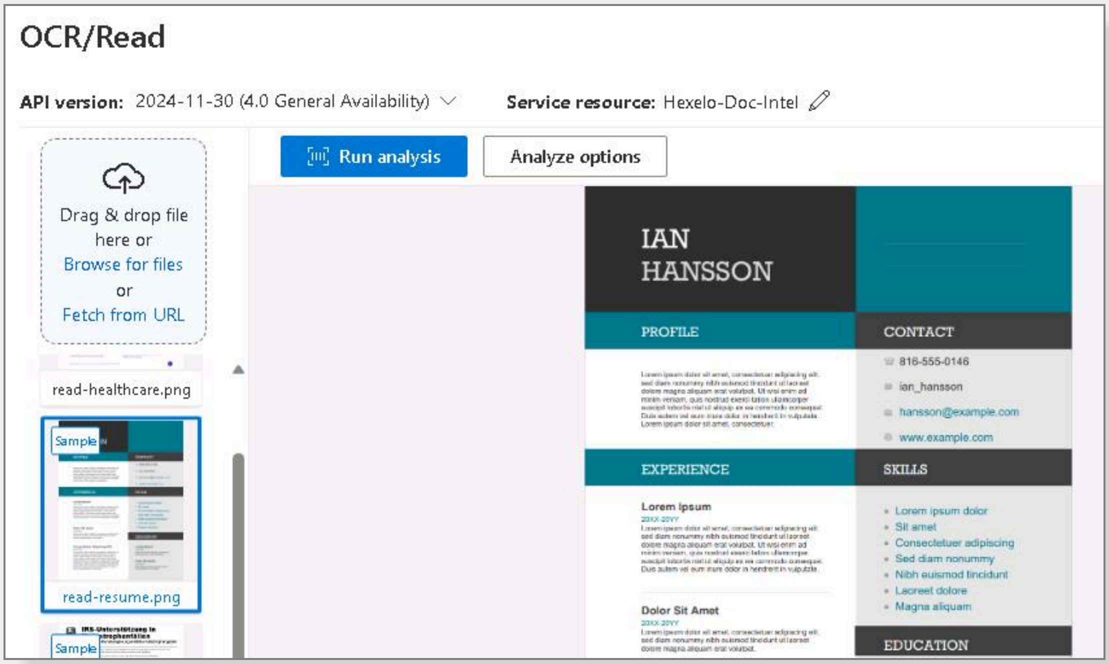
Want to learn more? Review the following documentation:

- Configuring a service resource
- Document Intelligence read model

- Analyze the **read-resume.png** sample image and then extract printed text by saving the JSON result to the T D:\LabFiles folder.

Expand this hint for guidance on extracting printed text from an image.

- On the Document Intelligence Studio | OCR/Read page, in the left pane, select the **read-resume.png** sample file.



- Select **Run analysis**.

- Using your mouse, hover over the bounding boxes to view the contents listed within.

- In the right pane, on the Content tab, scroll down to review the content.
- Select the **Result** tab, and then select **Download** icon.

- In the Downloads dialog, hover over **read-resume.png.json**, and then select the **Show in folder** button.

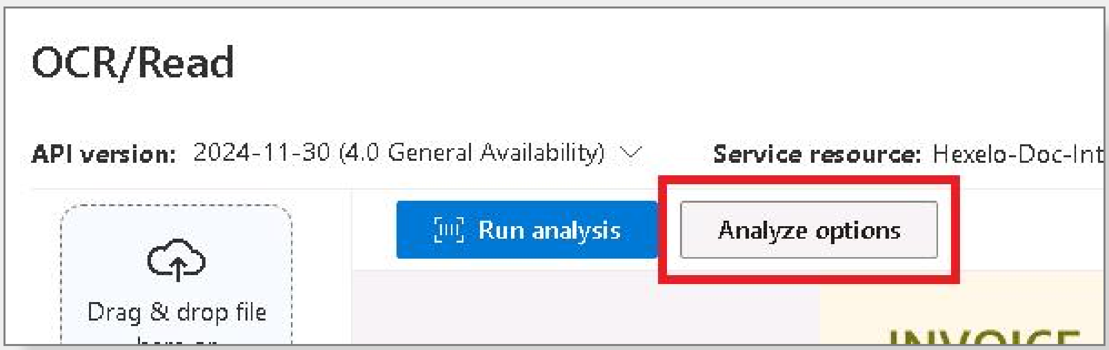
- In the Downloads window, move the **read-resume.png.json** file to the **D:\LabFiles** folder.

- Analyze and read barcodes by using the **read-barcode.pdf** sample and then save the PDF result to the T D:\LabFiles folder.

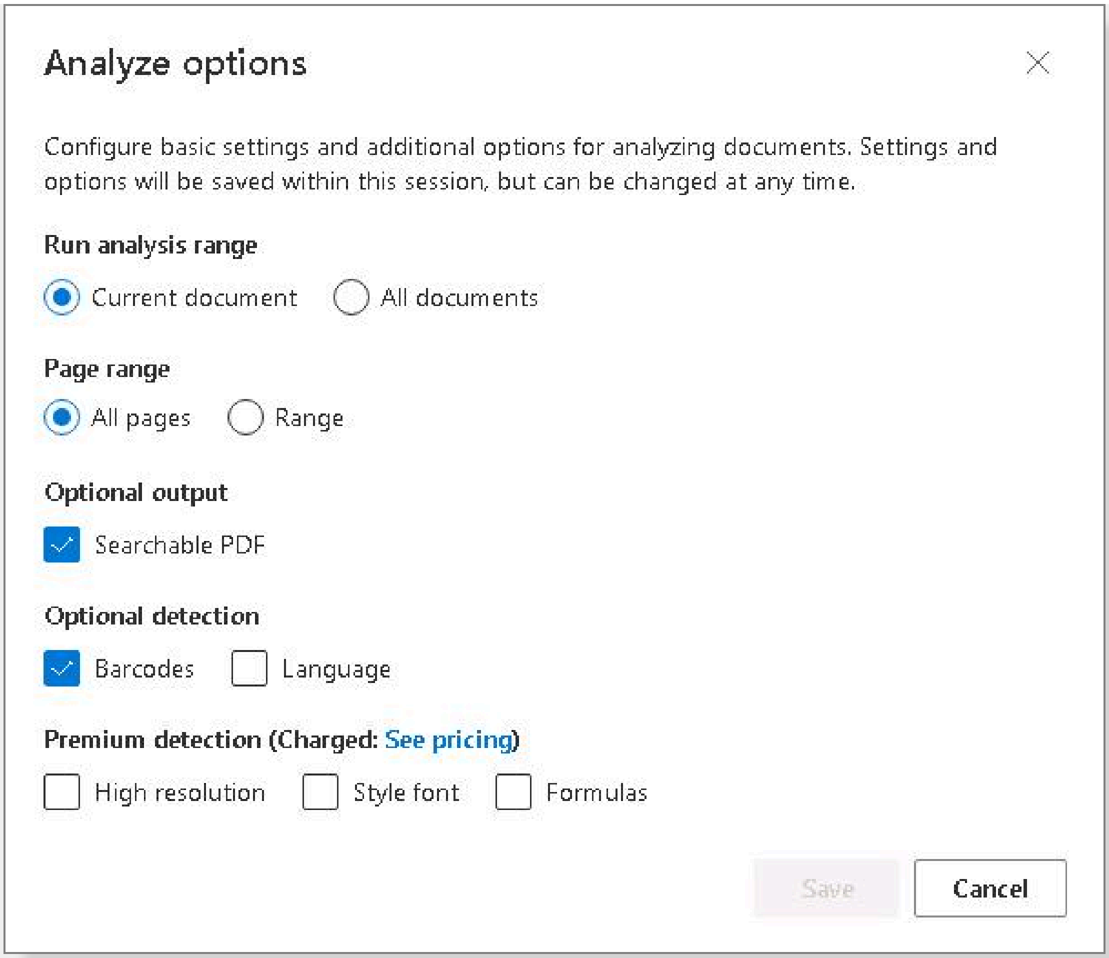
Expand this hint for guidance on analyzing and reading barcodes.

- On the Document Intelligence Studio | OCR/Read page, in the left pane, select the **read-barcode.pdf** sample file.

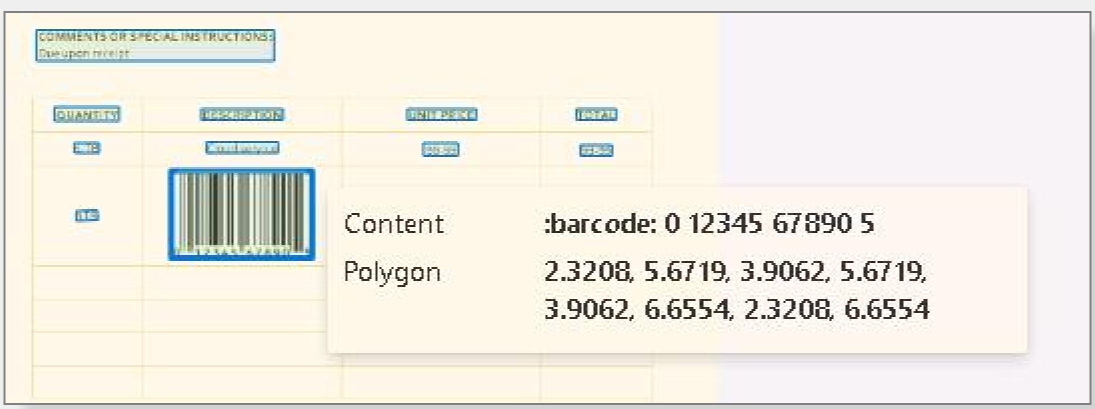
- Select **Analyze options**.



- In the Analyze options dialog, in Run analysis range, select **Current document**.
- In Page range, select **All pages**, and then in Optional output, select the **Searchable PDF** checkbox.
- In Optional detection, select the **Barcodes** checkbox, and then select **Save**.

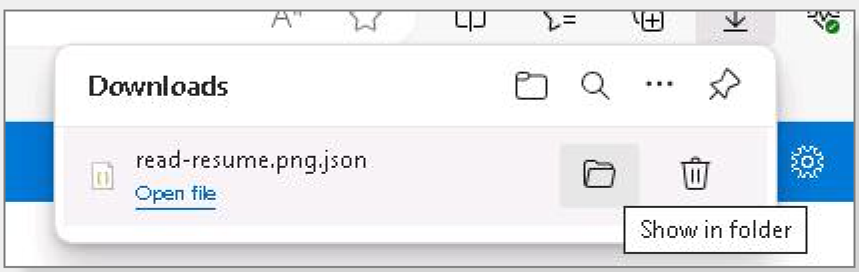


- Select **Run analysis**.
- Using your mouse, hover over the Barcode bounding box to view the contents listed within.



- In the right pane, select the **Content** tab.
- On the Content page, select the **Barcodes** tab, and then review the content.

- On the Content page, select the **Download** button.
- In the Downloads dialog, to the right of read-barcode.pdf, select the **Show in folder** button.

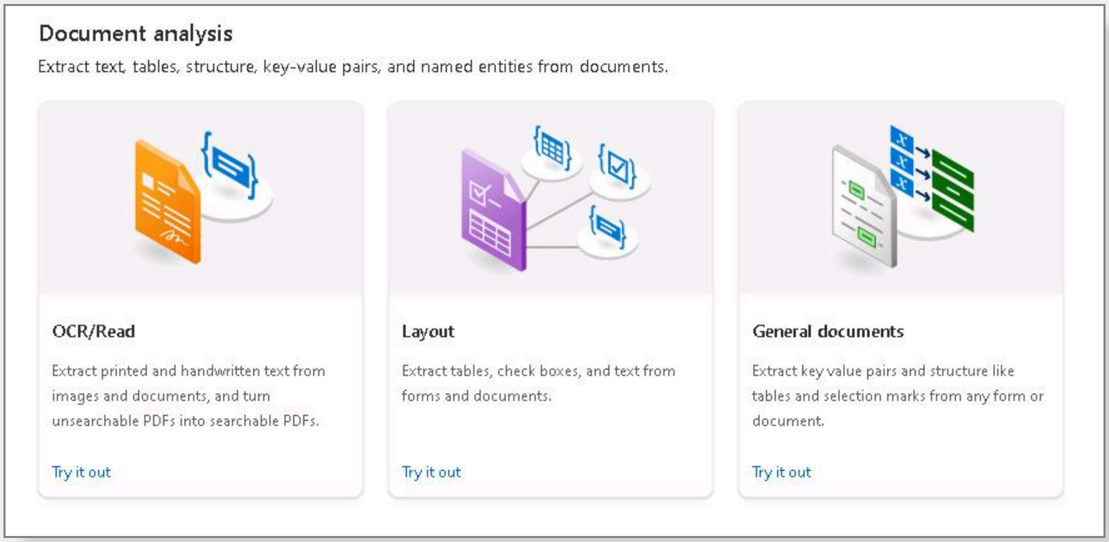


- In the Downloads folder, move the **read-barcode.pdf** file to the **D:\LabFiles** folder.

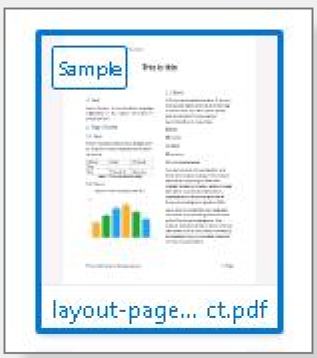
- Analyze the **layout-pageobject.pdf** sample file in the **Layout** model, and then extract tables, checkboxes, and text by saving the **JavaScript** code in the **D:\LabFiles** folder.

Expand this hint for guidance on analyzing a Layout model.

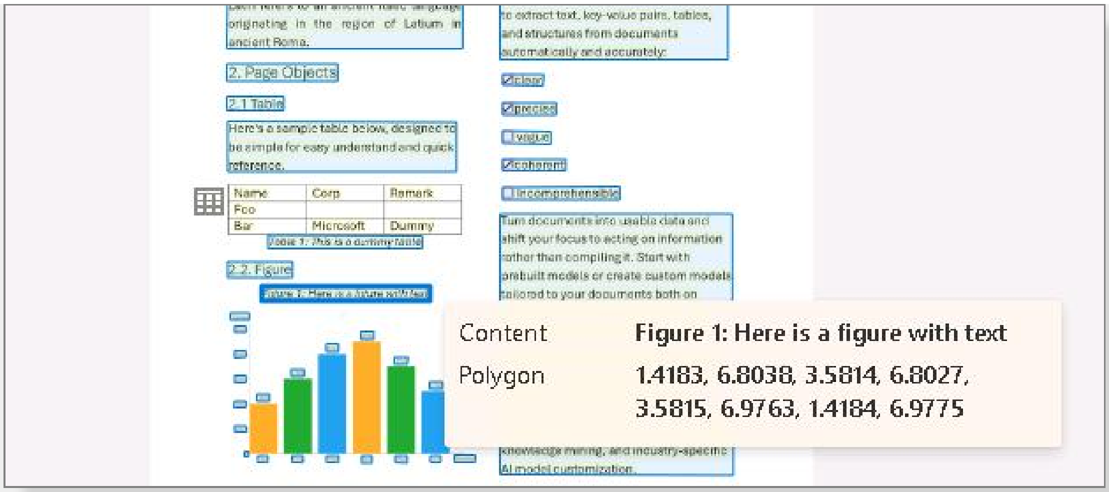
- On the breadcrumb menu, select **Document Intelligence Studio**.
- On the Document Intelligence Studio home page, in Document analysis, in **Layout**, select **Try it out**, and then dismiss all prompts.



- On the Layout page, in the left pane, select the **layout-pageobject.pdf** sample file.



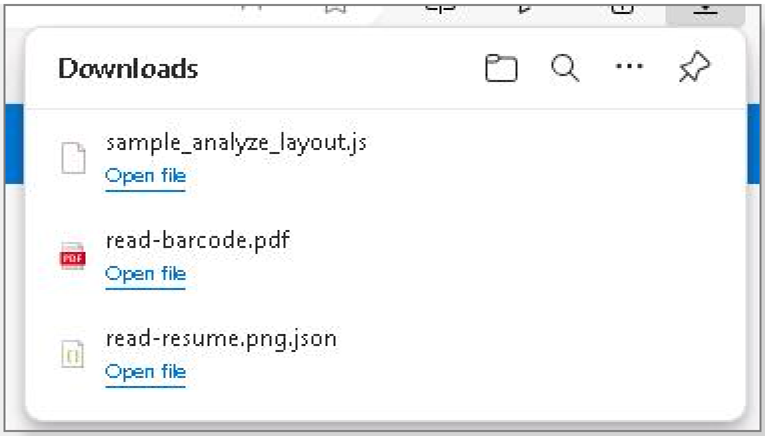
- Select **Run analysis**.
- Using your mouse, hover over the bounding boxes to view the contents listed within.



- In the right pane, on the Content tab, scroll down to review the content.
- Select the **Tables** tab, and then by using your mouse, hover over the **Table 1** diagram and notice the corresponding popup on the main document.

- Select the **Code** tab.
- In the language dropdown menu, select **JavaScript**, and then select the **Download** button.

- In the Downloads dialog, when prompted to Keep or delete the download, select **Keep**.
- Hover over **sample_analyze_layout.js** file by using your mouse, and then select the **Show in folder** button.



- In the Downloads folder, move the **sample_analyze_layout.js** file to the **D:\LabFiles** folder.

Want to learn more? Review the documentation on the Document Intelligence layout model.

Check your work

Verify

Create a Custom Extraction Model

Hints Enabled

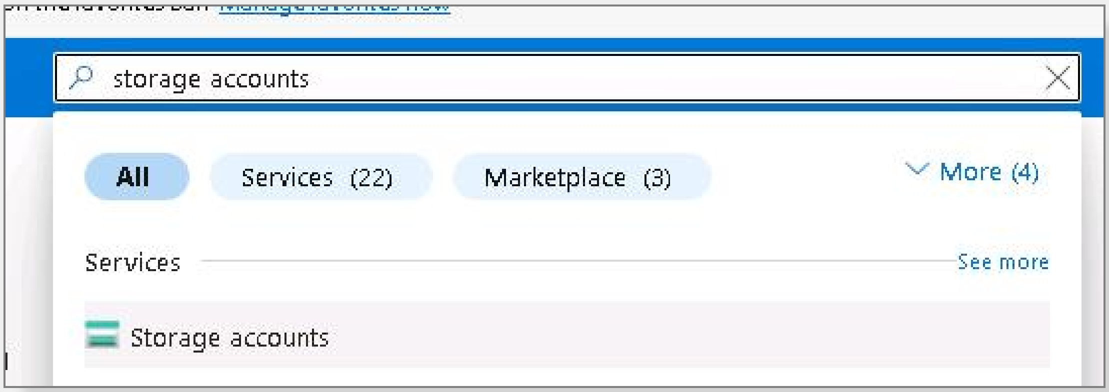
No Yes

- Open a new **Microsoft Edge** browser tab, go to T https://github.com/LODSContent/ChallengeLabs_Resources/blob/master/AI102/Samples.zip, save **samples.zip** to the T D:\LabFiles folder, and then extract the contents to the **D:\LabFiles\Samples** folder.
- Create an Azure T storage account by using the values in the following table. For any property that is not specified, use the default value.

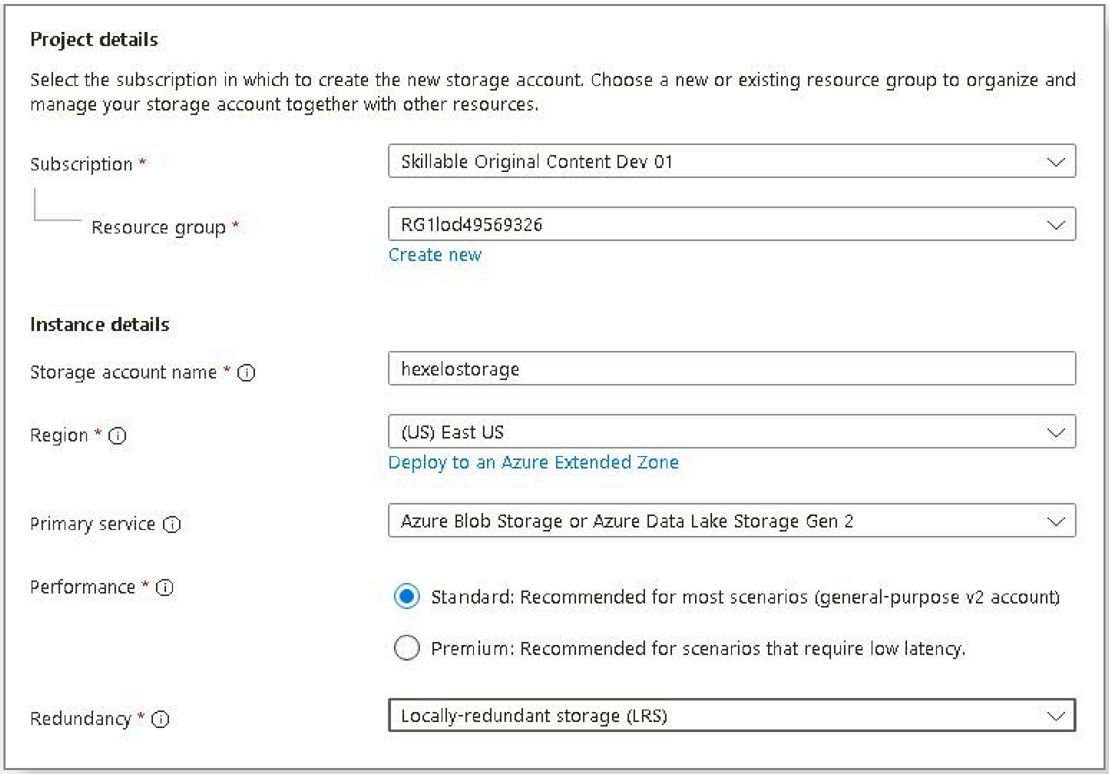
Property	Value
Resource group	**{RESOURCE_GROUP_NAME}**
Name	T hexelostorage{LAB_INSTANCE_ID}
Region	**East US**
Primary service	**Azure Blob Storage or Azure Data Lake Storage Gen 2**
Redundancy	**Locally-redundant storage (LRS)**

Expand this hint for guidance on creating a storage account.

- On the **Microsoft Azure** browser tab, in the upper-left corner, select the **show portal menu** icon, and then select **Create a resource**.
- On the Create a resource blade, in *Search resources, services, and docs (G+/)*, enter storage account, and then select **Storage accounts**.



- On the Marketplace blade, select the **Storage account** tile.
- On the Storage account blade, select **Create**.
- On the Create a storage account blade, on the Basics page, in Project details, in Resource group, select **{RESOURCE_GROUP_NAME}**.
- In Instance details, in Storage account name, enter hexelostorage{LAB_INSTANCE_ID} .
- In Region, select **(US) East US**.
- In Primary service, select **Azure Blob Storage or Azure Data Lake Storage Gen 2**.
- In Performance, select **Standard: Recommended for scenarios (general-purpose v2 account)**.
- In Redundancy, select **Locally-redundant storage (LRS)**.



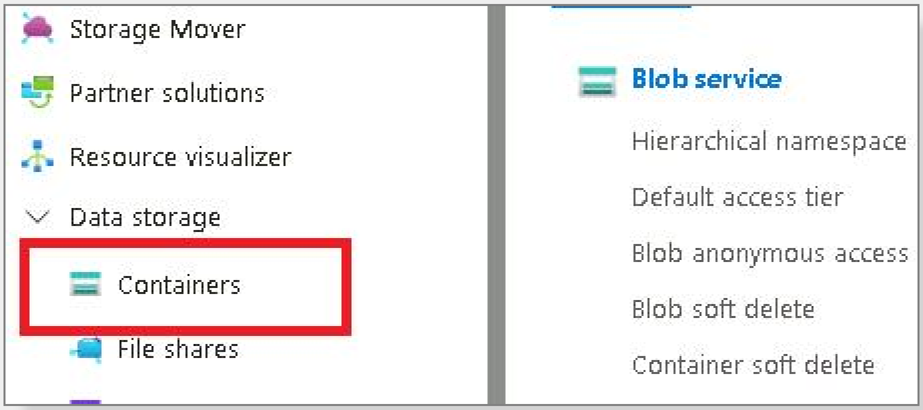
- On the Create a storage account blade, select **Review + create**, and then select **Create**.

Want to learn more? Review the documentation on creating an Azure storage account.

- Create a Blob container named T trainingsamples in the **hexelostorage{LAB_INSTANCE_ID}** storage account.

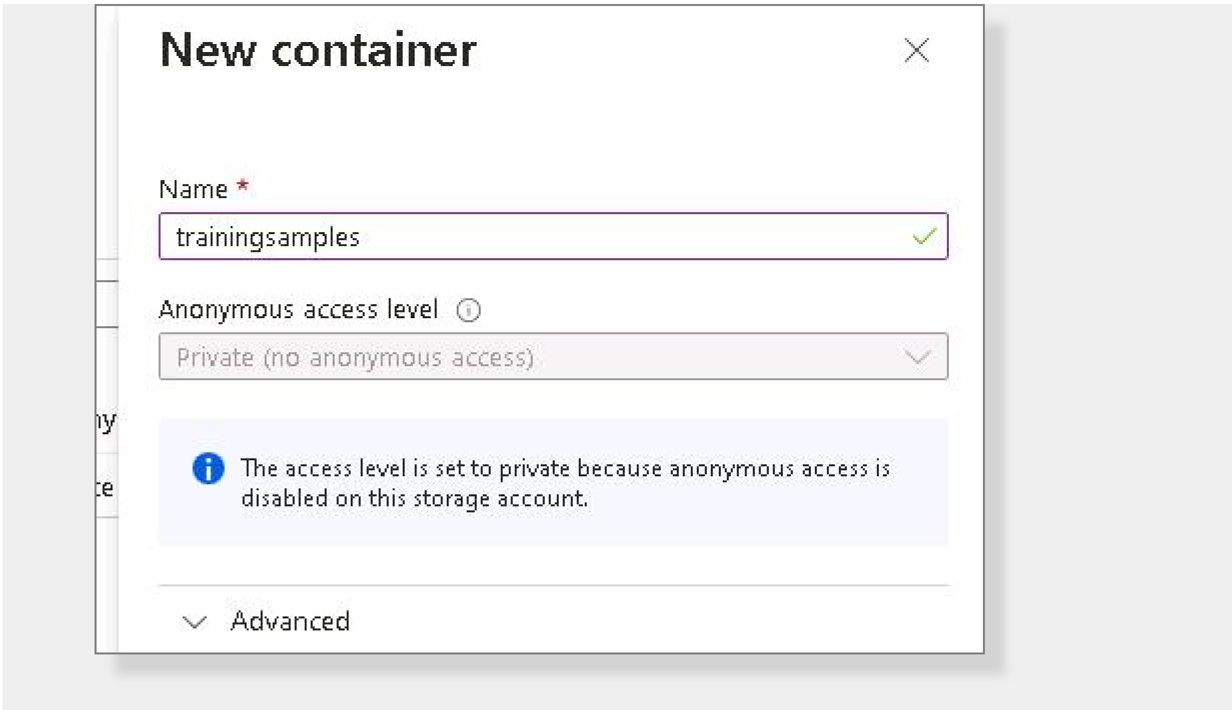
Expand this hint for guidance on creating a Blob container.

- On the Deployment page, select **Go to resource**.
- On the hexelostorage{LAB_INSTANCE_ID} service menu, in Data storage, select **Containers**.



- On the Containers page, on the command bar, select **+ Container**.

- On the New container blade, in Name, enter T trainingsamples, and then select **Create**.



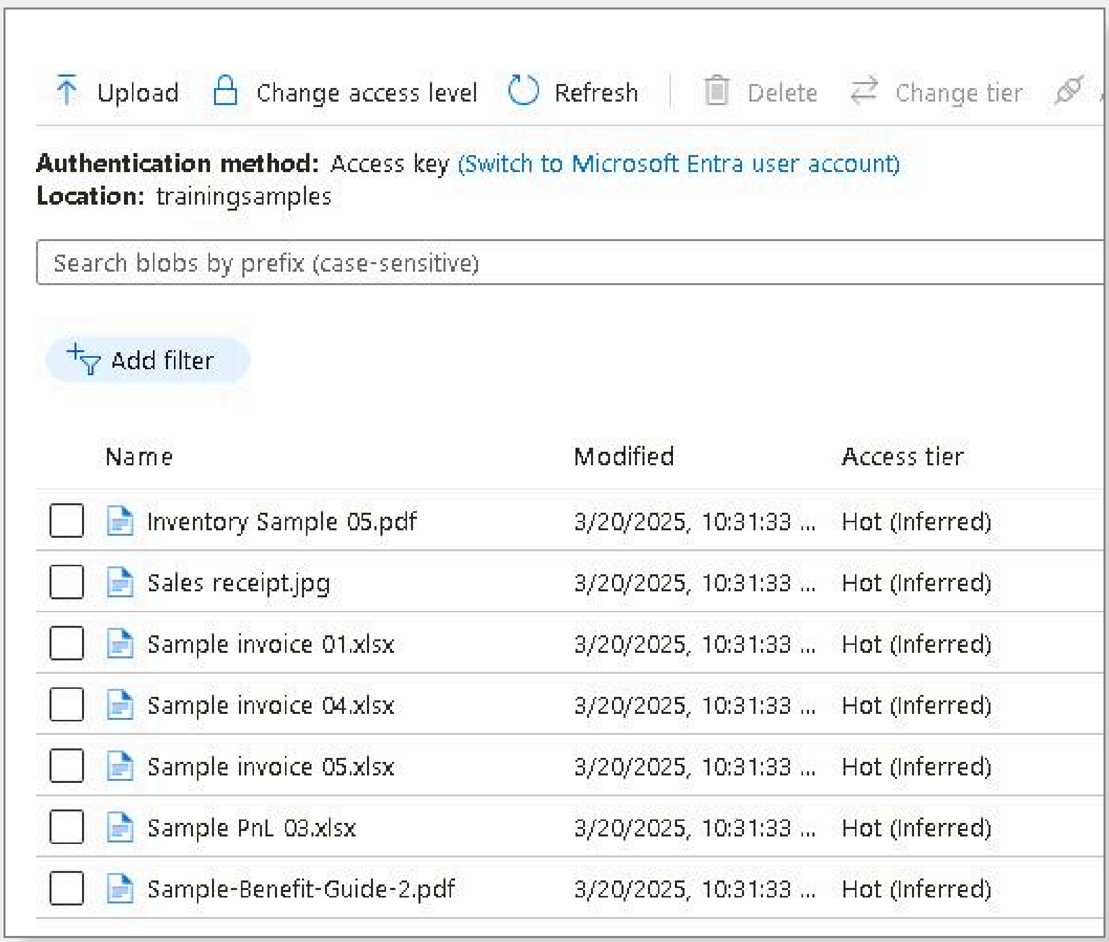
Want to learn more? Review the documentation on creating a Blob container.

- Upload the extracted sample.zip documents to the **trainingsamples** Blob container.

Expand this hint for guidance on uploading files to a Blob container.

- On the Containers page, select the **trainingsamples** container.

- On the trainingsamples blade, on the command bar, select **Upload**.
- On the Upload blob blade, select **Browse for files**.
- In the Open dialog, in the left pane, browse to the T D:\LabFiles\Samples folder, select all of the extracted documents, and then select **Open**.
- On the Upload blob blade, select **Upload**.

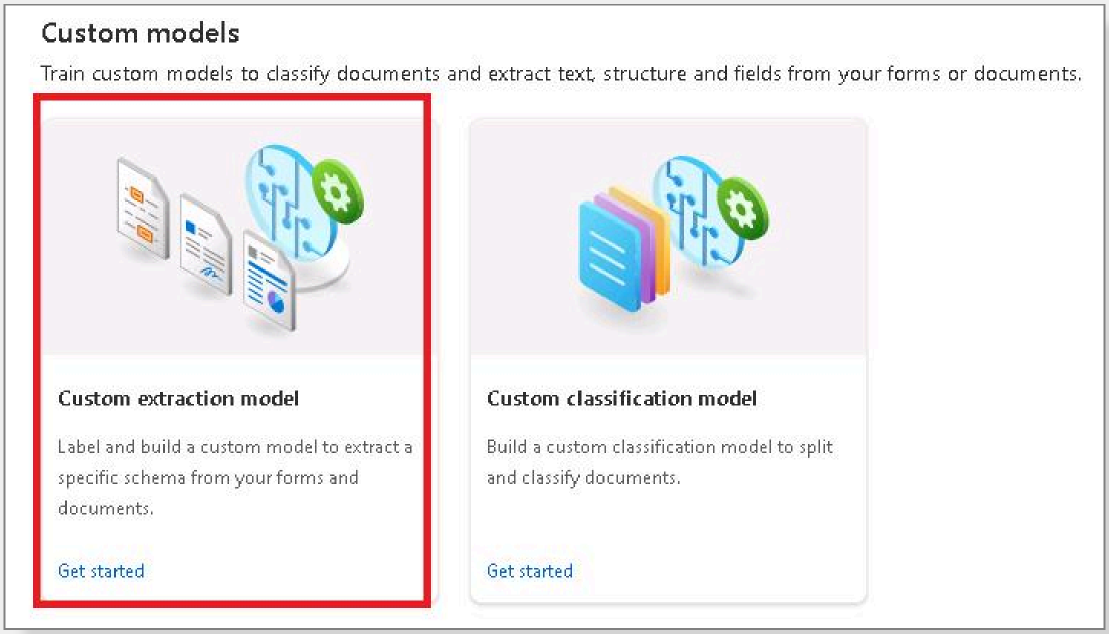


Want to learn more? Review the documentation on uploading files to a Blob container.

- Create a custom extraction model.

Expand this hint for guidance on creating a custom model.

- Select the **Document Intelligence Studio** browser tab.
- On the breadcrumb menu, select **Document Intelligence Studio**.
- On the Document Intelligence Studio home page, in Custom models, in Custom extraction model, select **Get started**.



- If needed, sign in as T {USERNAME} using T {PASSWORD} as the password.

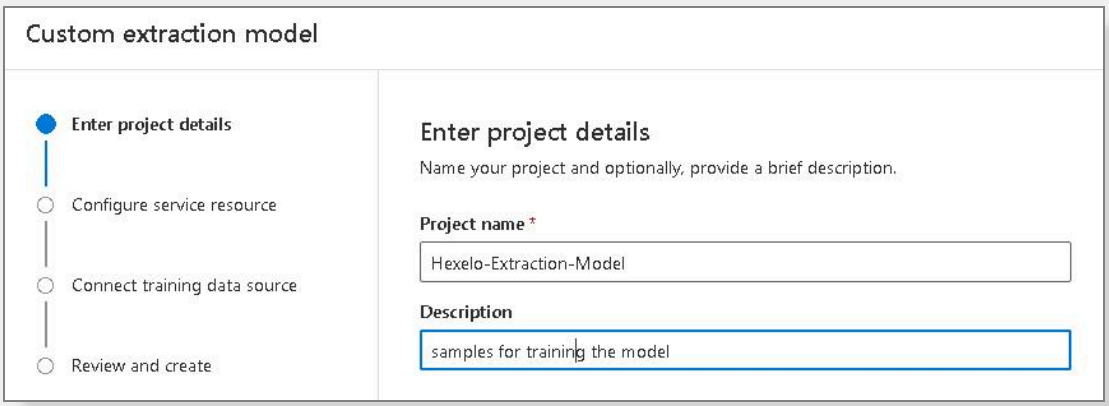
Want to learn more? Review the documentation on creating a custom model.

- Create a project named T Hexelo-Extraction-Model that has a description of T samples for training the model.

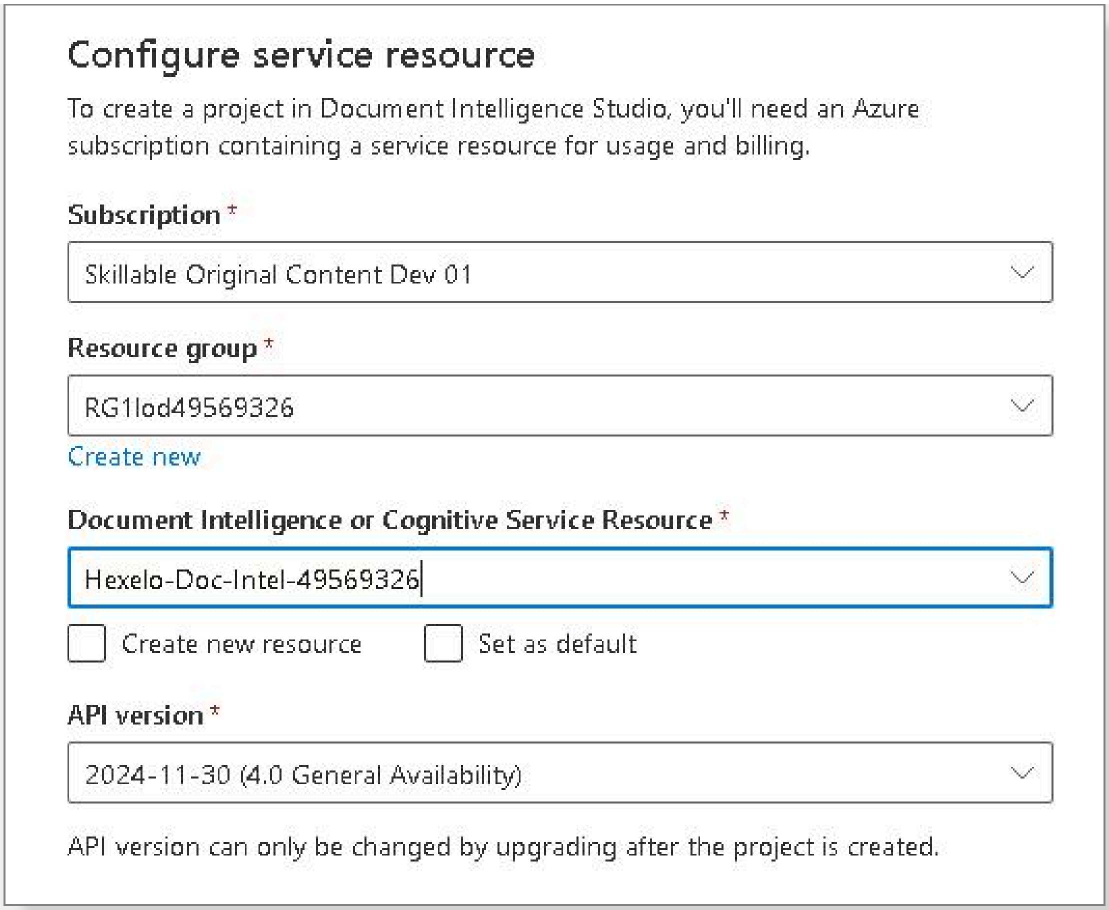
Expand this hint for guidance on creating a project.

- On the Custom extraction models page, in My Projects, select **+ Create a project**.

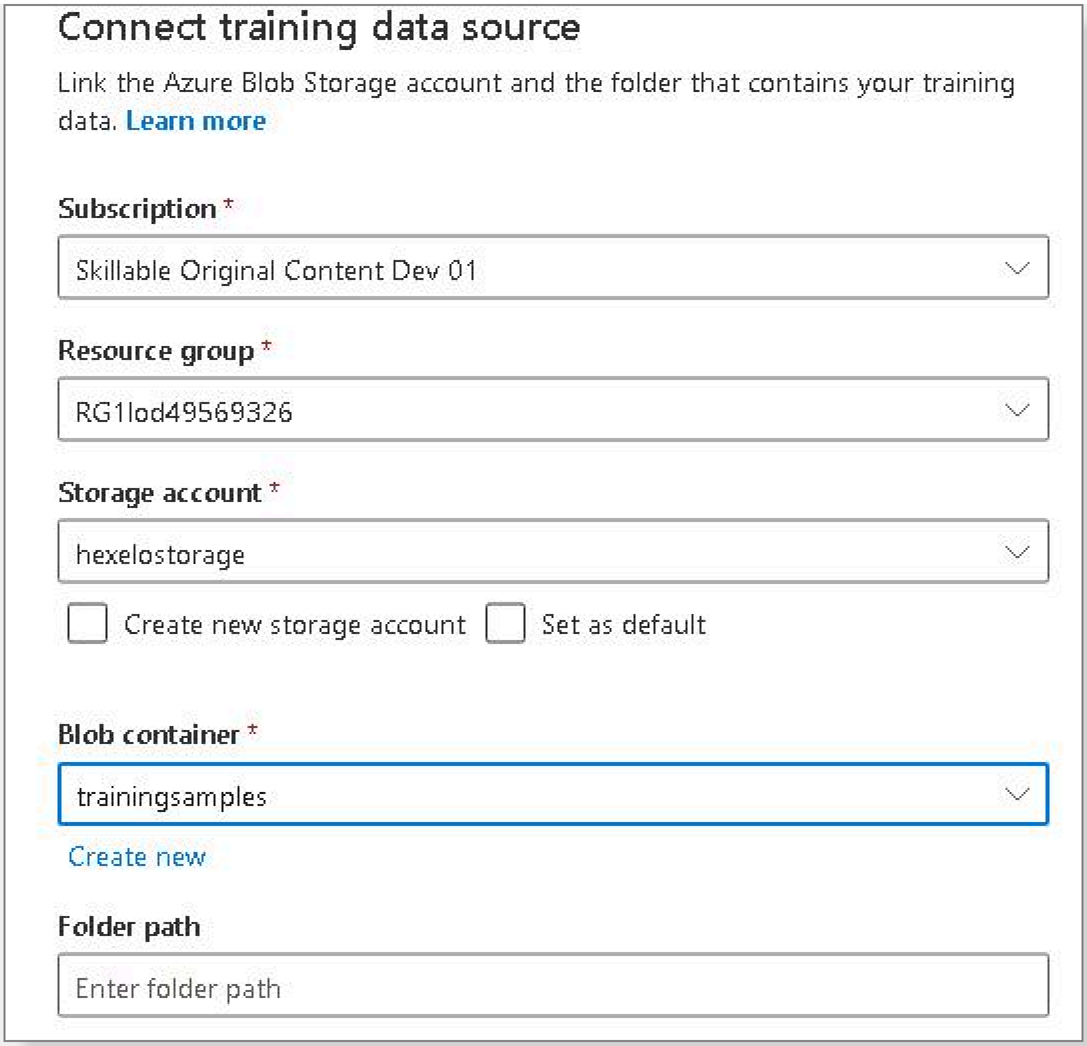
- In the Custom extraction model dialog, in Enter project details, in Project name, enter T Hexelo-Extraction-Model.
- In Description, enter T samples for training the model, and then select **Continue**.



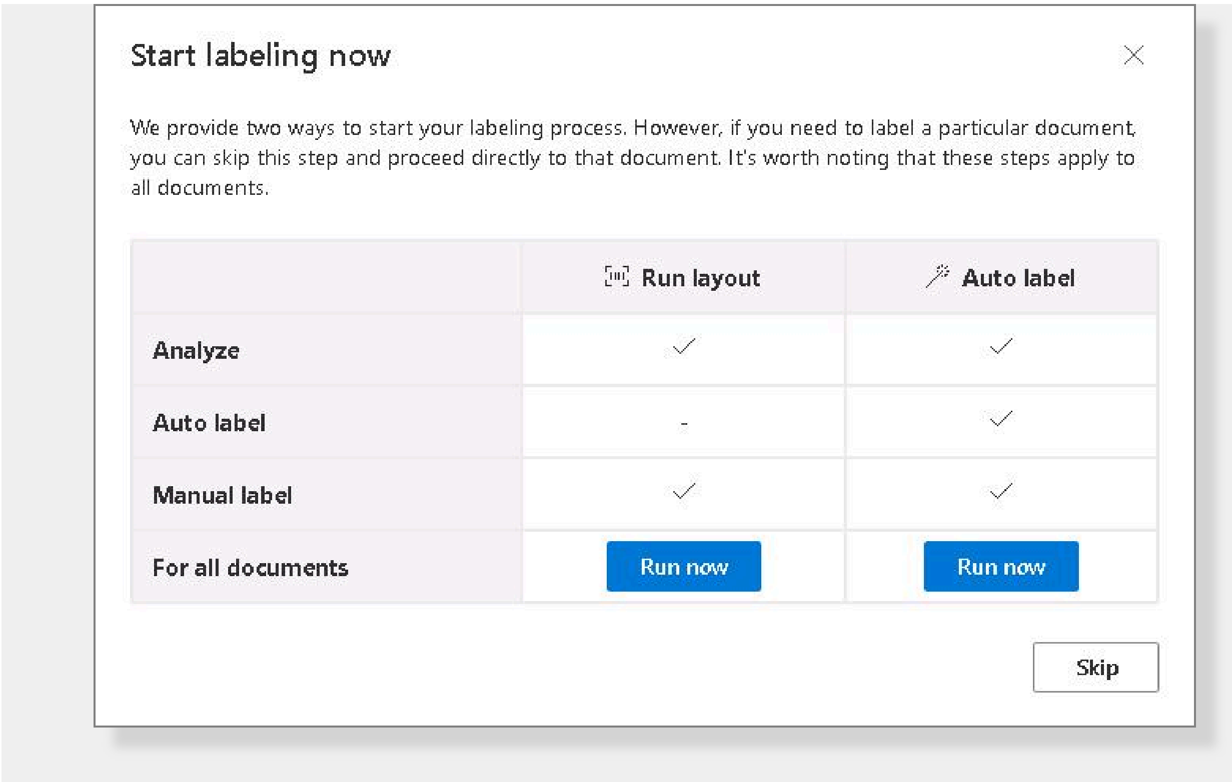
- On the Configure service resource page, in Subscription, select the default subscription-for example, **Skillable Original Content Dev 01**.
- In Resource group, select **{RESOURCE_GROUP_NAME}**.
- In Document Intelligence or Cognitive Service Resource, select **Hexelo-Doc-Intel-{LAB_INSTANCE_ID}**, and then select **Continue**.



- On the Connect training data source page, in Subscription, select the default subscription, and then in Resource group, select **{RESOURCE_GROUP_NAME}**.
- In Storage account, select **hexelostorage{LAB_INSTANCE_ID}**.
- In Blob container, select **trainingsamples**, and then select **Continue**.



- On the Review and create page, review the selections, and then select **Create project**.
- In the Start labeling now dialog, select **Skip**.



- You will verify the new project and model after you have trained it in the next exercise.

Want to learn more? Review the documentation on creating a project.

Check your work

Verify

Train a Custom extraction model

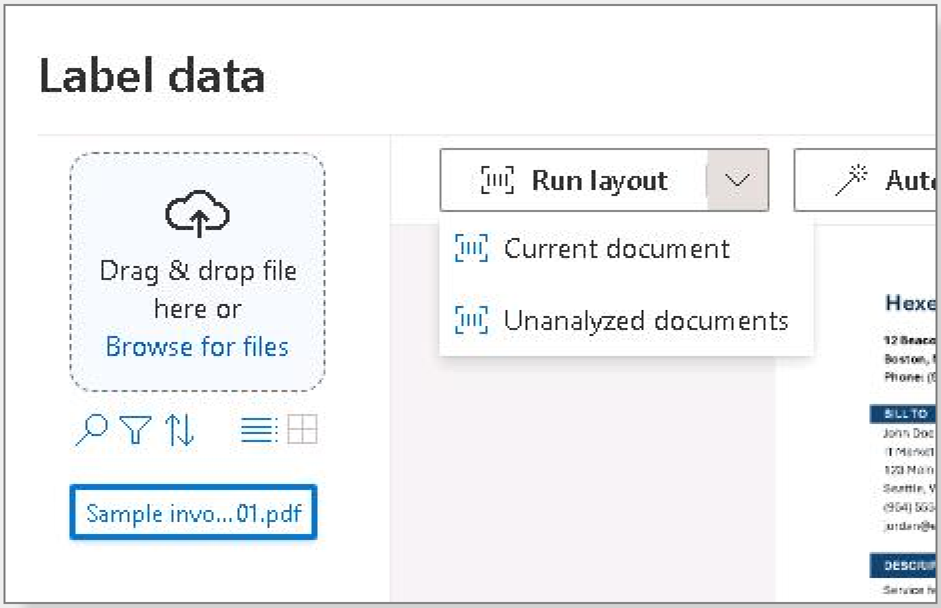
Hints Enabled

No Yes

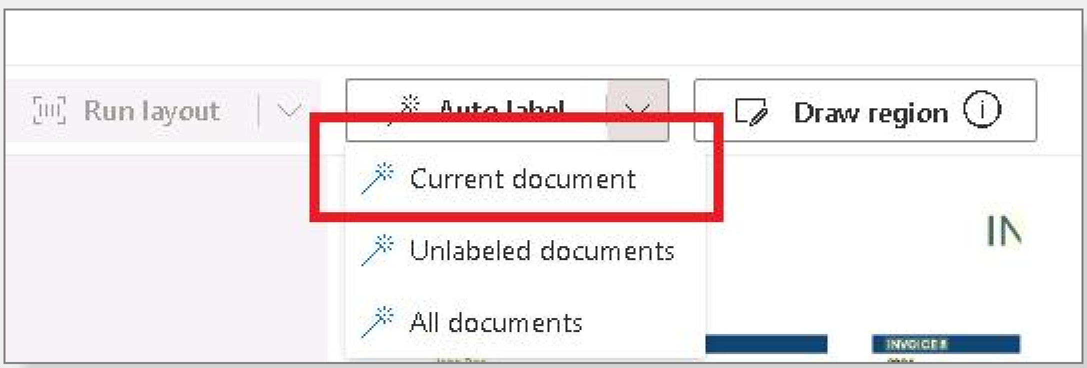
- Auto label **Sample invoice 01.pdf**.

Expand this hint for guidance on labeling data.

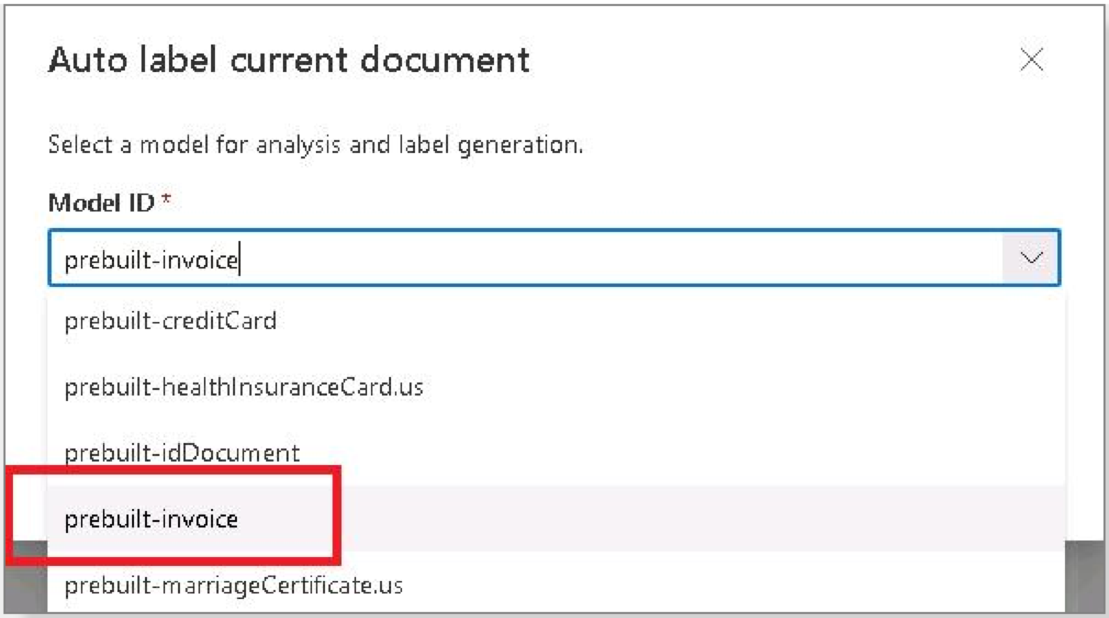
- On the Label data page, in the documents list, select **Sample invoice 01.pdf**, select **Run layout**, and then select **Current document**.



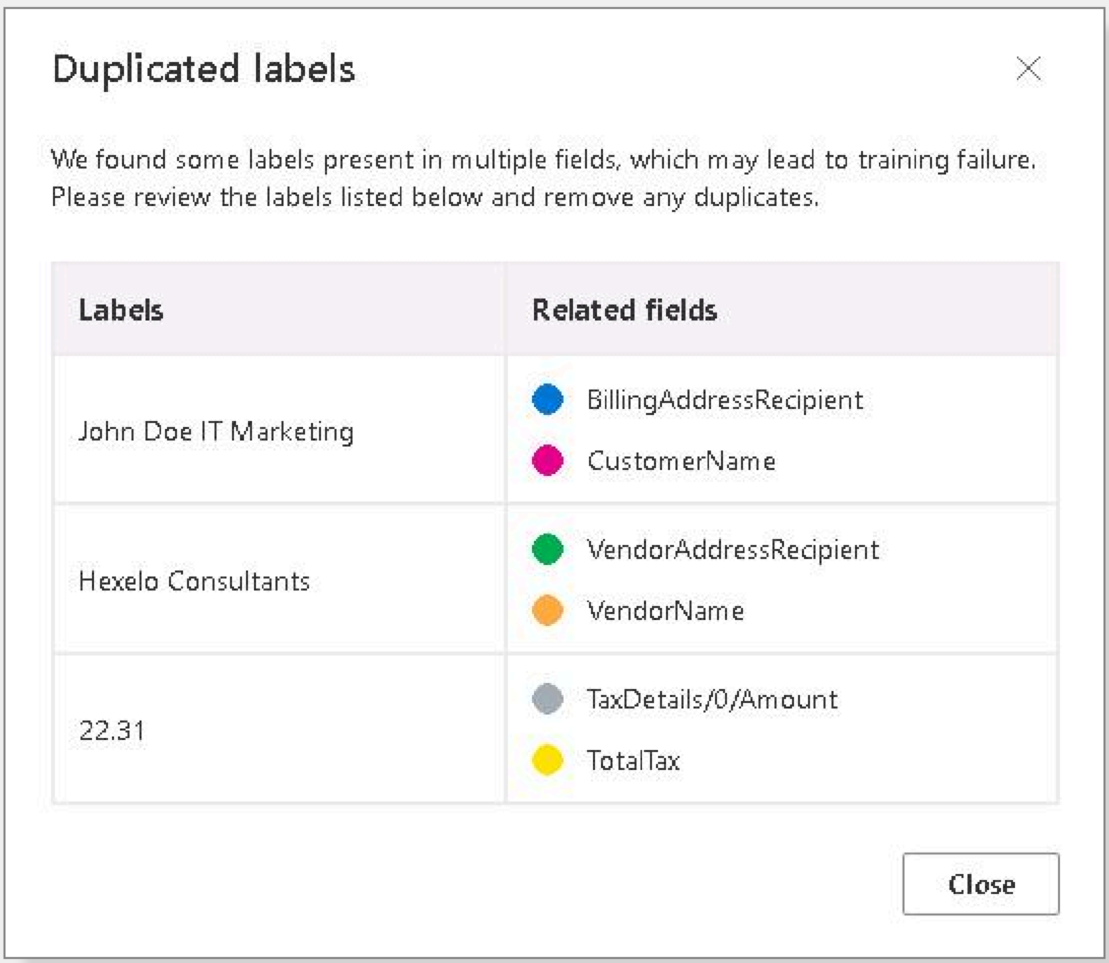
- Select **Auto label**, and then select **Current document**.



- In the Auto label current document dialog, in Model ID, select **prebuilt-invoice**, select the **I understand that existing labels will be ...** check box, and then select **Auto label**.



- In the Duplicated labels dialog, review the duplicates, and then select **Close**.

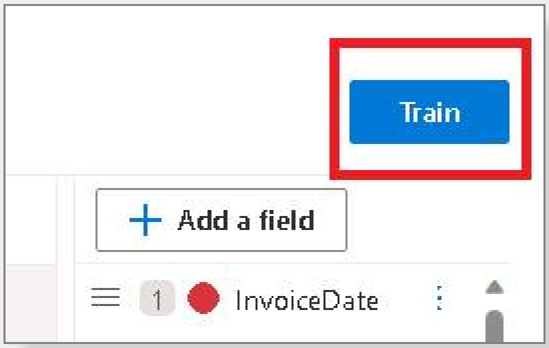


Want to learn more? Review the documentation on labeling a model.

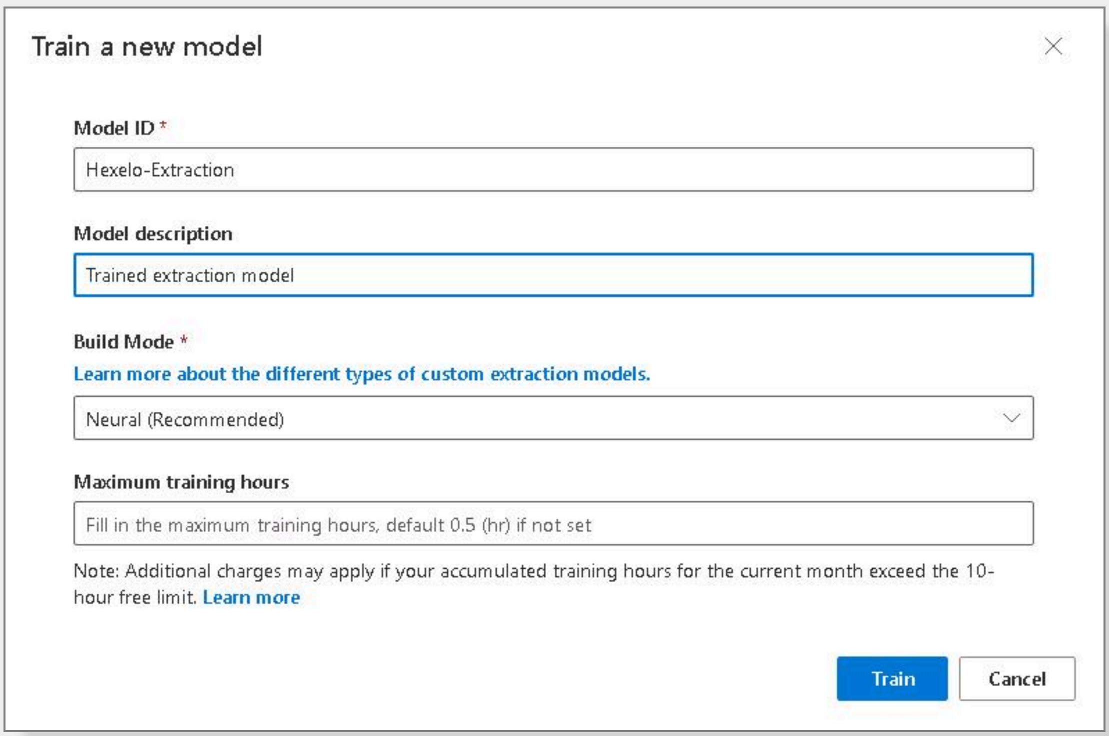
- Train a new model named T Hexelo-Extraction, that has a description of T Trained extraction model and a **Neural** Build mode, and then display the model.

Expand this hint for guidance on training a model.

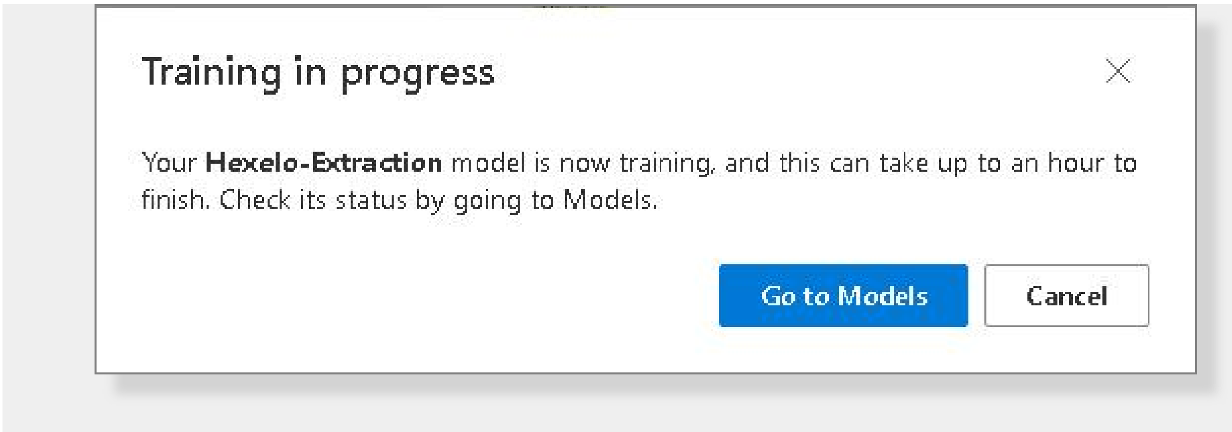
- On the Label data page, select the **Train** button.



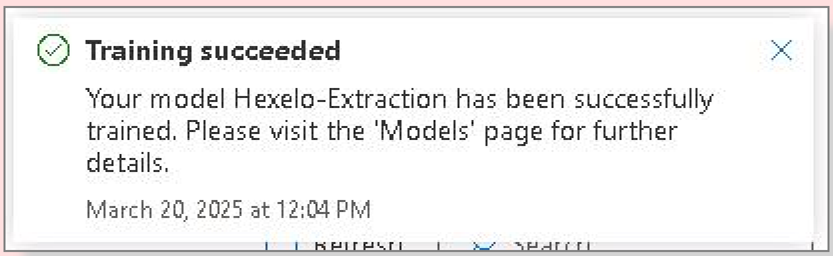
- In the Train a new model dialog, in Model ID, enter T Hexelo-Extraction.
- In Model description, enter T Trained extraction model.
- In Build mode, select **Neural (Recommended)**, and then select **Train**.



- In the Training in progress dialog, select **Go to Models** to view the model.



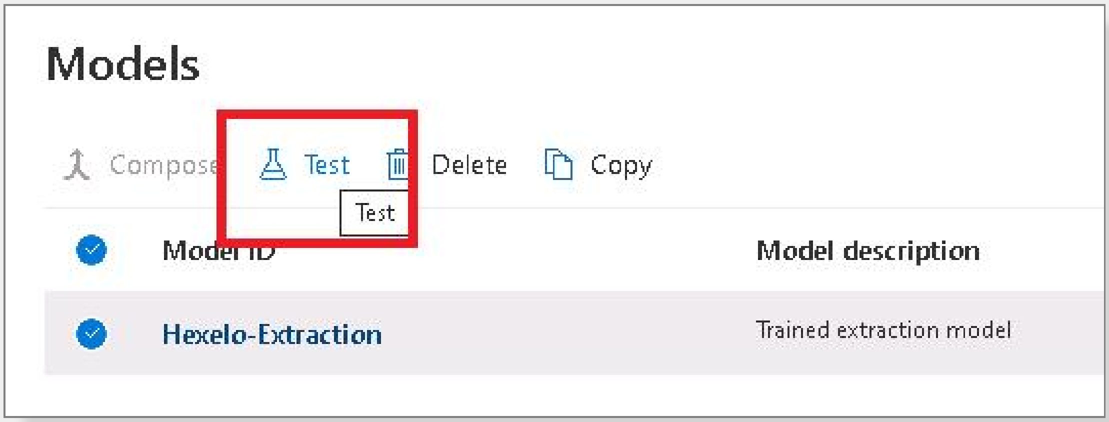
⚠ **Wait for the status to change from *running* to *succeeded* before continuing. It may take up to 5 minutes.**



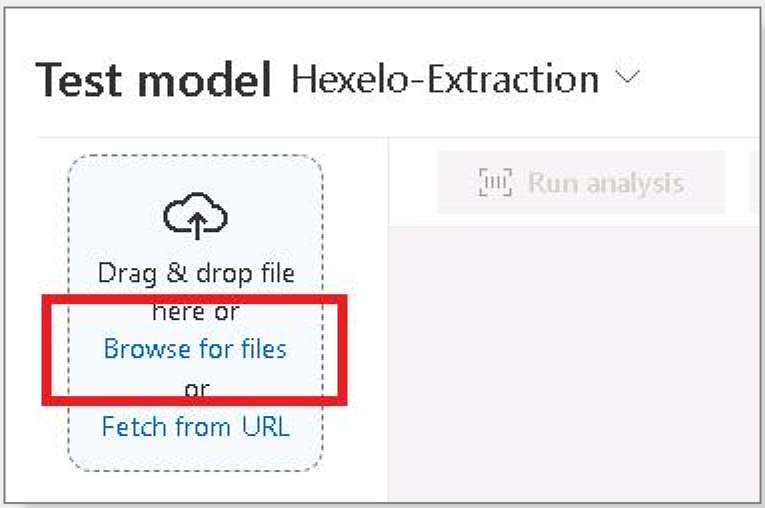
- **Test** the **Hexelo-Extraction** model on Sample invoice 04.pdf and then upload the sample to the dataset to assist in further training the model.

Expand this hint for guidance on testing a model.

- On the Models page, select the **Hexelo-Extraction** model, and then on the command bar, select **Test**.

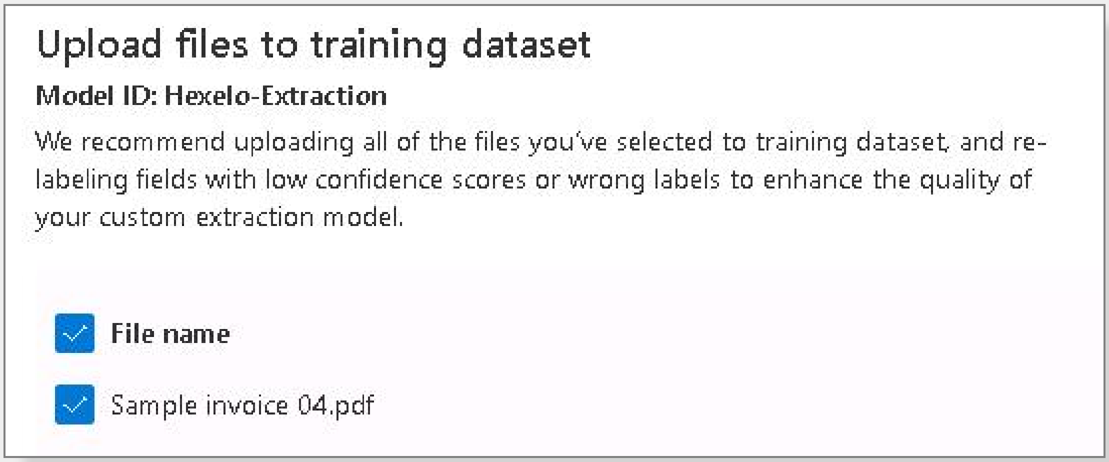


- On the Test model Hexelo-Extraction page, select **Browse for files**.
- In the Open dialog, select **Sample invoice 04**, and then select **Open**.

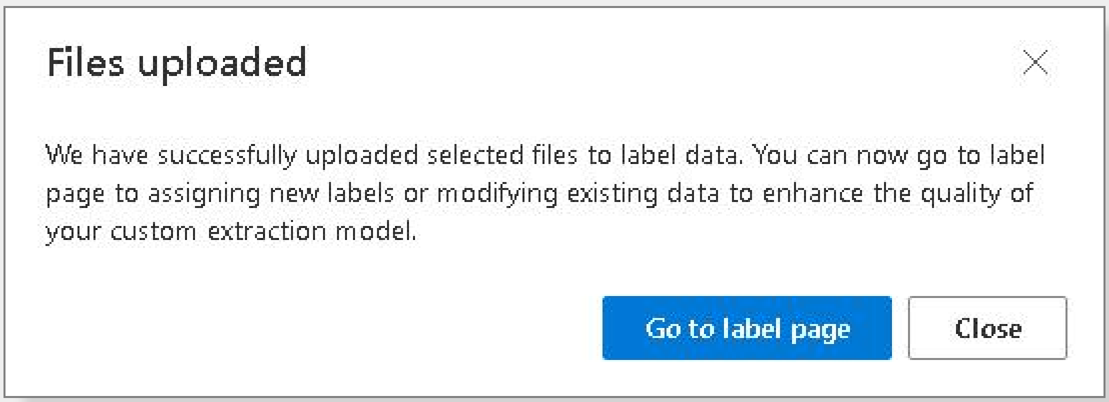


- On the Test model Hexelo-Extraction page, select **Run analysis**, and then review all of the Fields that were not found.
- Select **Mark as to be uploaded to training dataset**, and then select **Upload**.

- On the Upload files to training dataset blade, verify the file name, and then select **Upload files**.



- Select **Go to label page**.



You can continue to train your model to improve it by adding other invoices and documents to the data set.

- Delete the **Hexelo-Extraction** model.

Expand this hint for guidance on deleting the model.

- In the Custom extraction model pane, select **Models**.
- On the Models page, select **Hexelo-Extraction**, and then on the command bar, select **Delete**.

- Delete all Azure resources.

Expand this hint for guidance on deleting all Azure resources.

- On the Microsoft Azure portal browser tab, select the **Show portal menu** icon, in FAVORITES, select **All resources**, and then dismiss all prompts.
- On the All resources blade, select the **Name** checkbox to select all resources.
- On the command bar, select **Delete**.
- On the Delete Resources blade, enter T delete, and then select **Delete**.

Check your work

Verify

Summary

Congratulations, you have completed the **Explore Document Processing in Azure AI's Document Intelligence Studio** Challenge Lab.

You have accomplished the following:

- Set up and explored Document Intelligence Studio.
- Extracted and processed different types of data.
- Created a Custom extraction Model.
- Trained a Custom extraction model.

Ending your lab

To ensure your lab is recorded as complete, select **Submit** or **End** below. Exiting [X] the lab will result in an incomplete status.

⚠ Once you select **Submit** or **End**, you will not be able to return to this Challenge Lab.

Your feedback is important!

As you end your Challenge Lab, please take a few minutes to complete the short survey that will appear in the next window. Alternatively, you may provide your feedback directly to Challenge Labs feedback.

Looking for your next Challenge Lab?

Below are recommended Challenge Labs to try next.

- Explore Document Processing in Azure AI's Document Intelligence Studio [Guided]
- Create a Bot by Using the Bot Framework SDK [Guided]
- Generate Insights from Images by Using Azure AI Vision Services [Guided]
- Build an AI Model in Azure AI Machine Learning Studio [Guided]
- Integrate Azure AI Immersive Reader Into Your Application by Using Its SDK [Guided]
- Summarize Text Automatically by Using Azure AI Language Services [Guided]

- Translate Text Between Languages by Using Azure AI Translator [Guided]
- Generate Image Descriptions from Text by Using Azure AI Services [Guided]
- Build an AI-Powered Search System by Using Azure AI Search [Guided]
- Analyze and Generate Text-Based Sentiment by Using Azure AI Services [Guided]